



Beaverhill Bird Observatory: Building an End-to-End Prediction Model

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Agenda

1	Problem Framing & Business Understanding
2	Exploratory Data Analysis (EDA) Feature Engineering & Data Preparation
3	Modeling Approach & Key Decisions
4	Evaluation & Results
5	Streamlit / Deployment Concept Reflections, Limitations, and Next Steps

Problem

FRAMING / BUSINESS UNDERSTANDING

Understanding Owl Detectability Over Time

Classifying Owl Movement Behaviour Using ML

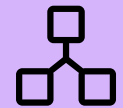
EDA

EXPLORATORY DATA ANALYSIS



Initial Data Shape

Initial data had 305845 rows and 45 columns, each row was one owl detection.



Missing Values & Cleaning

Filled the missing values by carrying over the closest value from above or below. `bfill().ffill()`

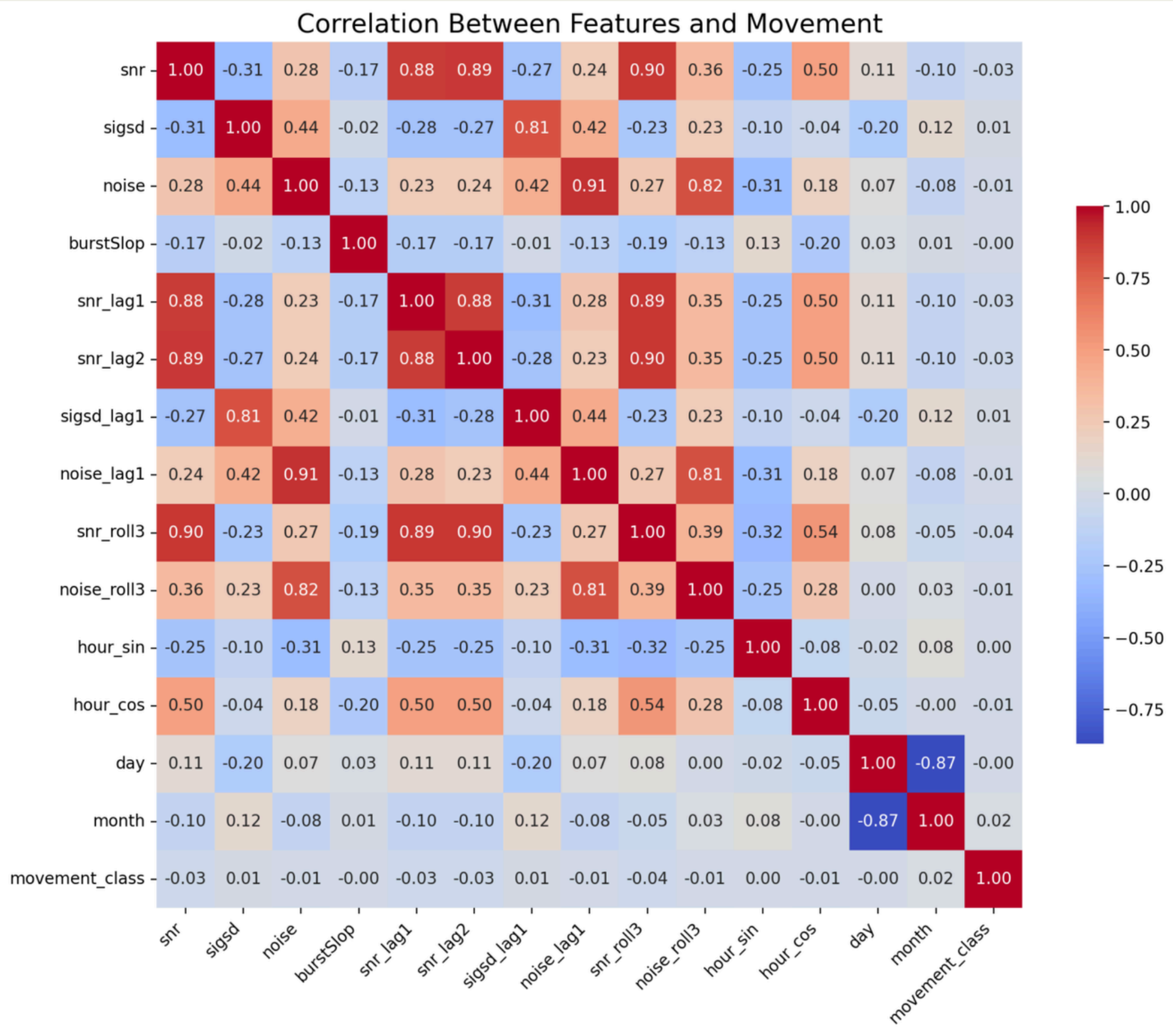


Final Cleaned Dataset

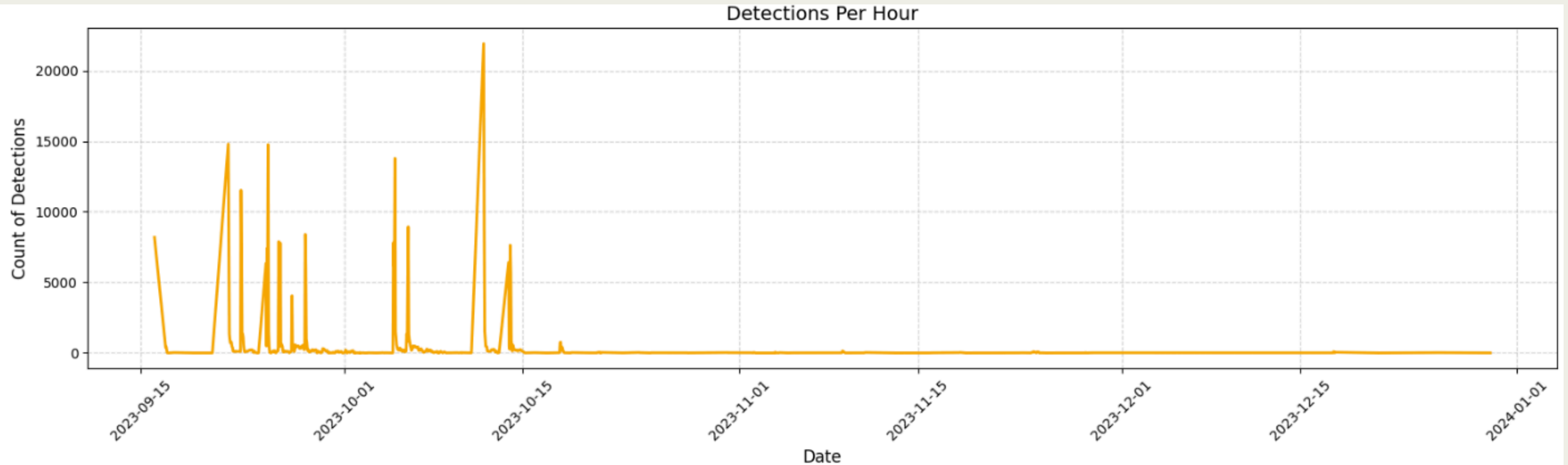
Final dataset kept only strong, complete records for better analysis.

Correlation between Feature & Movement

- Shows how owl detectability signals change over time.
- Helps us understand which features may influence movement behaviour.
- Highlights features that add value to the ML model.



Detections Per Hour



- Detection spikes show when owls are most active.
- Patterns separate resting vs. movement periods.
- These trends help ML classify owl behaviour accurately.

Feature Engineering

Creating Predictive Signals



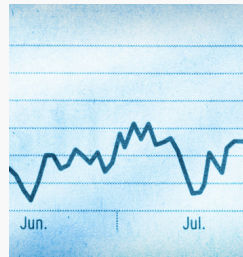
Time-based features

(hour, day, month, weekday) to capture owl activity patterns



Lag features

(previous signal & SNR readings) to include recent detection history



Rolling averages

to capture short-term signal trends

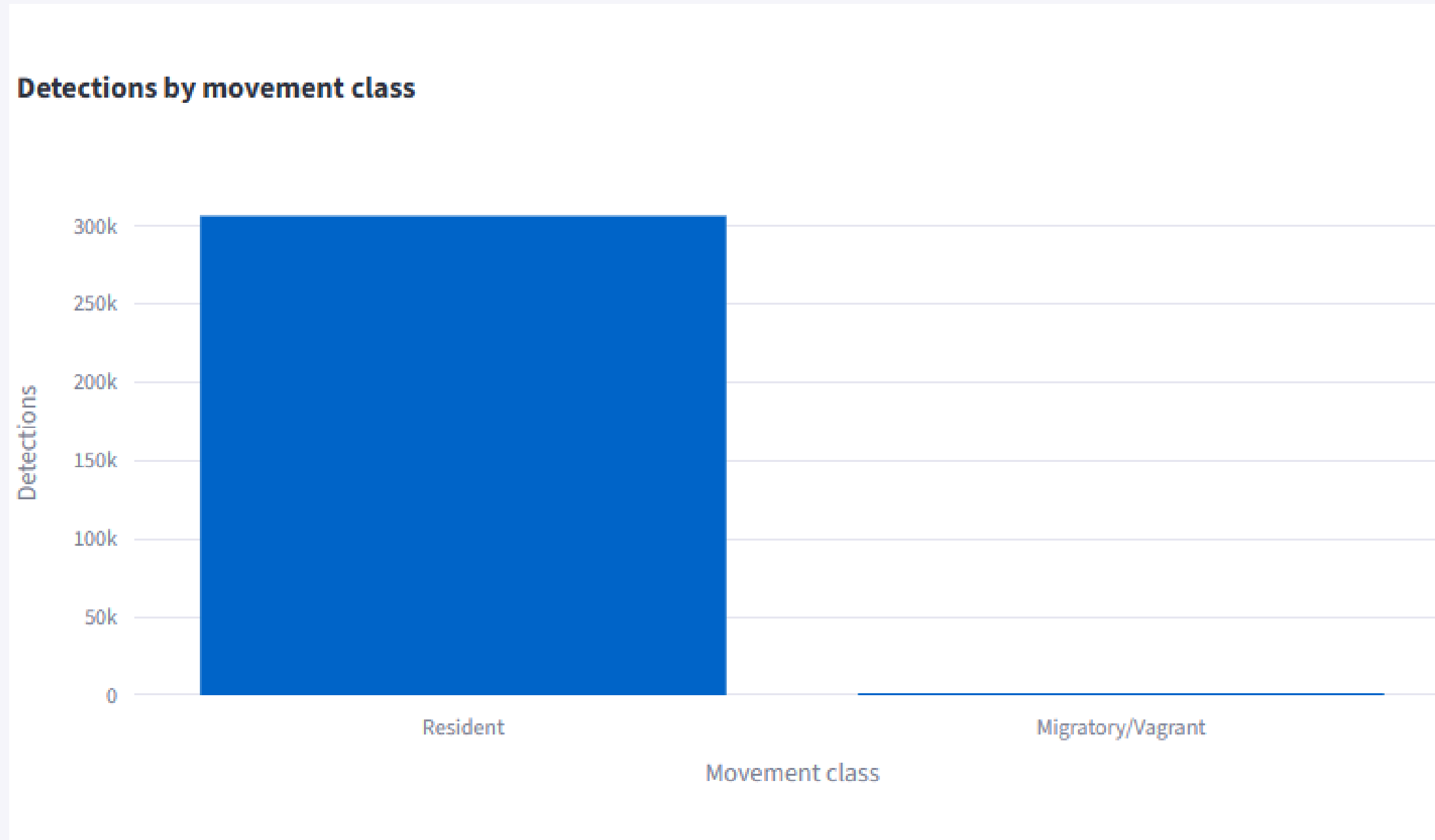


Behavioral flags

for weak, strong, or noisy signals



Movement Behaviour Classifier



1 Most detections belong to Resident owls, showing they remain within the monitored area for long periods.

2 Very few detections fall under Migratory/Vagrant behaviour, indicating rare movement events that are harder to observe.

3 This imbalance highlights the challenge of classifying owl movement behaviour and the need for ML models that can learn from limited migratory data

Modeling Approach & Key Decisions

Two prediction tasks:

- Classification: Predict owl movement behaviour type.



- Regression: Predict owl signal strength over time



Model used:

Random Forest Classifier
for movement prediction



XGBoost Classifier

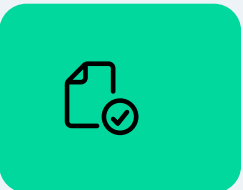


classification

Regression



Random Forest regressors
for signal strength



XGBoost Regressor

Evaluation Metrics

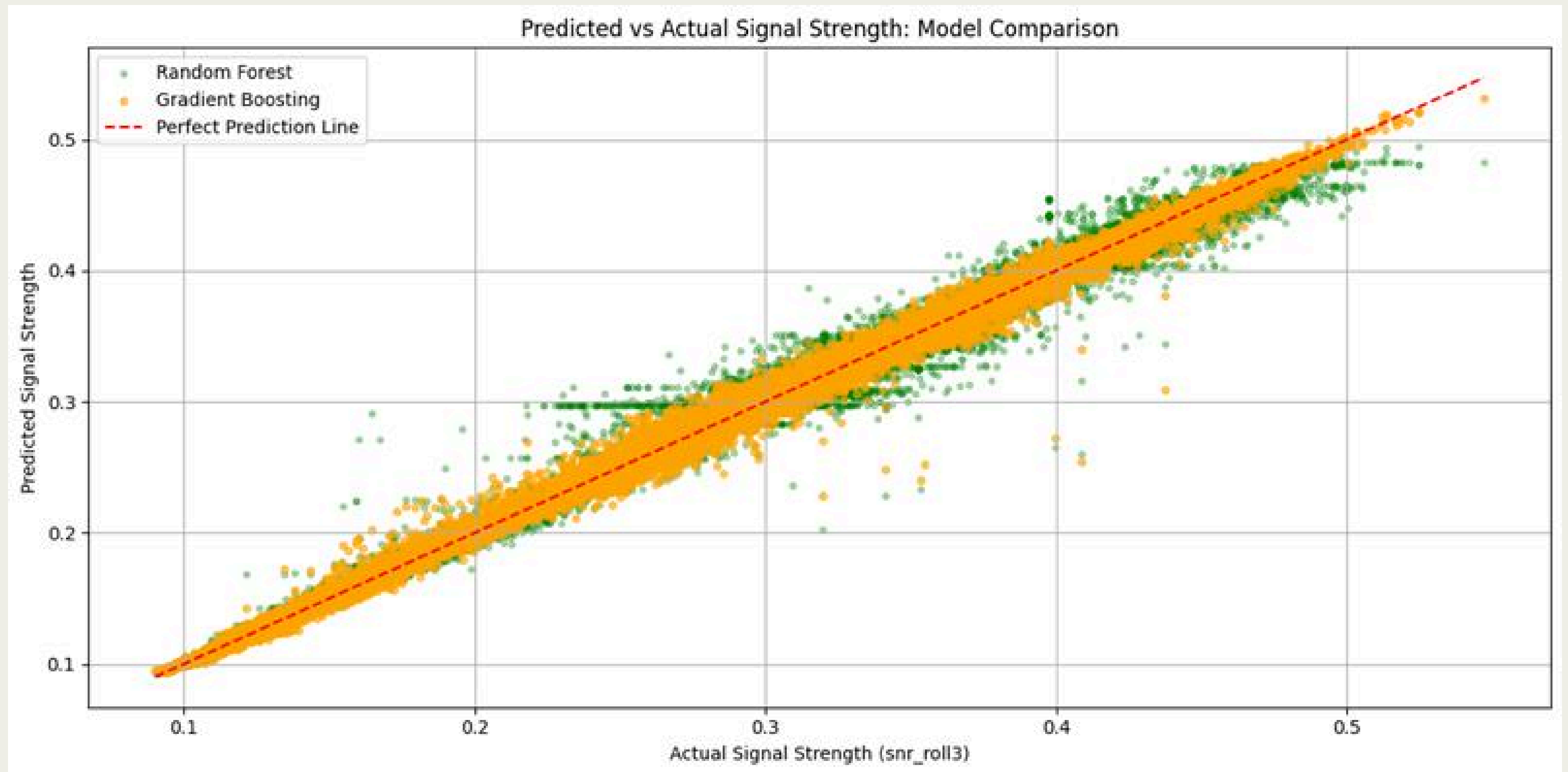
Regression Model Results

Model	RMSE	R2
Random Forest	0.006604	0.995408
Gradient Boosting	0.004281	0.998071

Classification Model Result

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
Random Forest	0.9995	0.9996	0.9995	0.9995
Gradient Boosting	1.0	1.0	1.0	1.0

Evaluation & Result



- *Tight clustering along the diagonal → strong model accuracy*
- *Gradient Boosting hugs the line more closely → best fit*
- *Very small spread → signal strength is highly predictable*

Streamlit Prototype & Deployment

Our team built an interactive Streamlit prototype that allows the Beaverhill Bird Observatory to:

- Explore owl detections over time
- Visualize modeled signal predictions
- View feature importance
- Compare actual vs predicted signal strength
- Ask questions using a simple chatbot interface

Streamlit Prototype

Reflections, Limitation & Next Step

Reflection

Our end-to-end ML pipeline successfully captured meaningful detection patterns.

Both the classifier and regression models learned real biological behavior, showing that owl detectability follows predictable temporal and signal-strength trends.

Next Step

Expand the dataset to include more owls, multiple receivers, and environmental data, and deploy a more scalable version of the Streamlit prototype to support real-time monitoring and early detection of migration or tag failure.

Limitation

Our dataset has uneven detection coverage across owls, limiting the model's ability to generalize movement patterns and capture rare migration events.



Thank you